



CSA0501-DATABASE MANAGEMENT SYSTEMS

LAB EXPERIMENTS

DATE: 14/07/2026

EXPERIMENT 9 – Query with Joins – EquiJoin, InnerJoin, OuterJoin

AIM:

To perform **JOIN operations** using **Equi Join, Inner Join, and Outer Join** on the given relations in the MySQL **milk_dairys** database and retrieve related records from multiple tables.

PROCEDURE:

- Open **MySQL** and create the database **milk_dairys**.
1. Select the database using:
2. USE **milk_dairys**;
3. Create the required tables with suitable attributes and constraints.
4. Insert sample records into the tables.
5. Perform an **Equi Join** by joining two tables using the equality (=) condition.
6. Perform an **Inner Join** to retrieve only the records that have matching values in both tables.
Perform an **Outer Join** to retrieve matching records along with unmatched records from one or both tables.
7. Execute each JOIN query and verify the output.
8. Compare the results obtained from Equi Join, Inner Join, and Outer Join.
9. Confirm that the JOIN operations are executed successfully..

PROGRAM:

1. CREATE TABLE

```
CREATE TABLE milk_dairys (  
  name VARCHAR(20),  
  liters INT,  
  amount INT,  
  account_number BIGINT,  
  phone_number BIGINT,  
  bank_name VARCHAR(20)  
); Then Insert Values:  
INSERT INTO milk_dairys(name, liters, amount, account_number, phone_number, bank_name)  
VALUES  
( 'Shanthi', 5, 320, 9679411723, 8639516519, 'SBI'),  
( 'Nagu', 4, 290, 4192233411, 7182111462, 'SBI'),  
( 'Sai', 7, 500, 497481234, 7670972570, 'SBI'),  
( 'Venu', 9, 650, 727446123, 9204639364, 'Canara'),
```

```
('Sugana', 10, 700, 825849623, 6305857568, 'Andhra'),
('Sekar', 8, 600, 889241729, 9180091111, 'SBI'),
('Kamala', 15, 1500, 6284942417, 9132277911, 'SBI'),
('Ramu', 11, 950, 723967941, 8918286963, 'SBI'),
('Sita', 9, 650, 5492411763, 9350005229, 'Andhra'),
('Laxmi', 4, 290, 46418234748, 891991835, 'Canara');
```

OUTPUT:

```
mysql> SELECT*FROM milk_dairys;
```

name	liters	amount	account_number	phone_number	bank_name
Shanthi	5	320	9679411723	8639516519	SBI
Nagu	4	290	4192233411	7182111462	SBI
Sai	7	500	497481234	7670972570	SBI
Venu	9	650	727446123	9204639364	Canara
Sugana	10	700	825849623	6305857568	Andhra
Sekar	8	600	889241729	9180091111	SBI
Kamala	15	1500	6284942417	9132277911	SBI
Ramu	11	950	723967941	8918286963	SBI
Sita	9	650	5492411763	9350005229	Andhra
Laxmi	4	290	46418234748	891991835	Canara

10 rows in set (0.00 sec)

CREATE 2ND TABLE:

```
create table bank_details(
name VARCHAR(50),
bank_name VARCHAR(50));
```

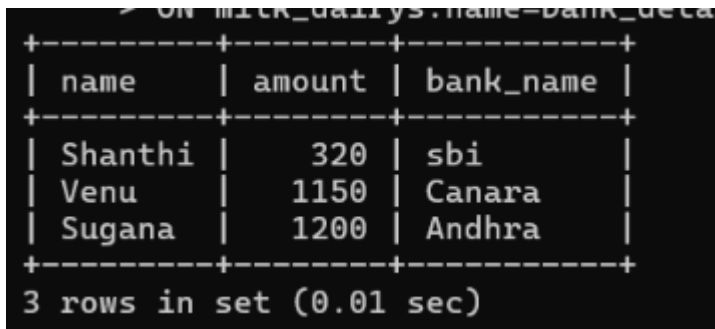
INSERTED VALUES:

```
INSERT INTO bank_details VALUES
('shanthi','sbi'),
('nagu','sbi'),
('sailu','sbi'),
('velu','canara'),
('suguna','andhra');
```

1)EQUAL JOIN

```
SELECT milk_dairys.name,
milk_dairys.amount,
bank_details.bank_name
FROM milk_dairys
JOIN bank_details
ON milk_dairys.name=bank_details.name;
```

OUTPUT:



```
SQL> ON milk_dairys.name=bank_details.name;
```

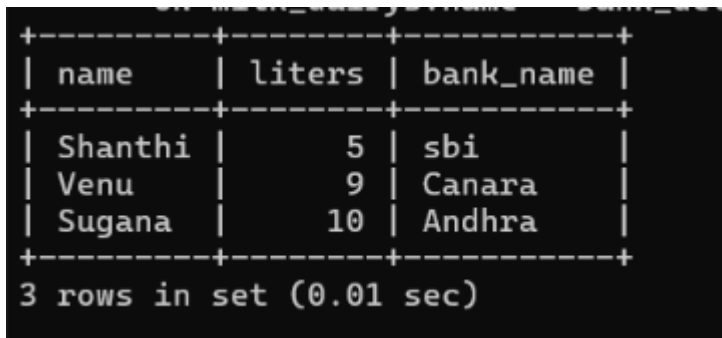
name	amount	bank_name
Shanthi	320	sbi
Venu	1150	Canara
Sugana	1200	Andhra

3 rows in set (0.01 sec)

2. INNER JOIN

```
SELECT milk_dairys.name, milk_dairys.liters, bank_details.bank_name  
FROM milk_dairys  
INNER JOIN bank_details  
ON milk_dairys.name = bank_details.name;
```

OUTPUT:



```
SQL> ON milk_dairys.name=bank_details.name;
```

name	liters	bank_name
Shanthi	5	sbi
Venu	9	Canara
Sugana	10	Andhra

3 rows in set (0.01 sec)

3)LEFT OUTER JOIN

```
SELECT milk_dairys.name,  
milk_dairys.amount,  
bank_details.bank_name  
FROM milk_dairys  
LEFT JOIN bank_details  
ON milk_dairys.name=bank_details.name;
```

OUTPUT:

name	amount	bank_name
Shanthi	320	sbi
Sai	500	NULL
Venu	1150	Canara
Sugana	1200	Andhra
Sekar	600	NULL
Kamala	2300	NULL
Ramu	1450	NULL
Sita	1150	NULL

8 rows in set (0.00 sec)

4. RIGHT OUTER JOIN

```
SELECT milk_dairys.name, milk_dairys.amount,
bank_details.bank_name
FROM milk_dairys
RIGHT JOIN bank_details
ON milk_dairys.name = bank_details.name;
```

OUTPUT:

name	amount	bank_name
Shanthi	320	sbi
NULL	NULL	sbi
NULL	NULL	sbi
NULL	NULL	canara
NULL	NULL	andhra
NULL	NULL	SBI
NULL	NULL	SBI
NULL	NULL	SBI
Venu	1150	Canara
Sugana	1200	Andhra

10 rows in set (0.01 sec)

5) CROSS JOIN

```
SELECT milk_dairys.name,
milk_dairys.amount,
bank_details.bank_name
FROM milk_dairys
CROSS JOIN bank_details;
```

OUTPUT:

-> CROSS JOIN bank_details;

name	amount	bank_name
Sita	1150	sbi
Ramu	1450	sbi
Kamala	2300	sbi
Sekar	600	sbi
Sugana	1200	sbi
Venu	1150	sbi
Sai	500	sbi
Shanthi	320	sbi
Sita	1150	sbi
Ramu	1450	sbi
Kamala	2300	sbi
Sekar	600	sbi
Sugana	1200	sbi
Venu	1150	sbi
Sai	500	sbi
Shanthi	320	sbi
Sita	1150	sbi
Ramu	1450	sbi
Kamala	2300	sbi
Sekar	600	sbi
Sugana	1200	sbi
Venu	1150	sbi
Sai	500	sbi
Shanthi	320	sbi
Sita	1150	canara
Ramu	1450	canara
Kamala	2300	canara
Sekar	600	canara
Sugana	1200	canara
Venu	1150	canara
Sai	500	canara

Shanthi	320	canara
Sita	1150	andhra
Ramu	1450	andhra
Kamala	2300	andhra
Sekar	600	andhra
Sugana	1200	andhra
Venu	1150	andhra
Sai	500	andhra
Shanthi	320	andhra
Sita	1150	SBI
Ramu	1450	SBI
Kamala	2300	SBI
Sekar	600	SBI
Sugana	1200	SBI
Venu	1150	SBI
Sai	500	SBI
Shanthi	320	SBI
Sita	1150	SBI
Ramu	1450	SBI
Kamala	2300	SBI
Sekar	600	SBI
Sugana	1200	SBI
Venu	1150	SBI
Sai	500	SBI
Shanthi	320	SBI
Sita	1150	SBI
Ramu	1450	SBI
Kamala	2300	SBI
Sekar	600	SBI
Sugana	1200	SBI
Venu	1150	SBI
Sai	500	SBI
Shanthi	320	SBI
Sita	1150	Canara
Ramu	1450	Canara
Kamala	2300	Canara

Sekar	600	Canara
Sugana	1200	Canara
Venu	1150	Canara
Sai	500	Canara
Shanthi	320	Canara
Sita	1150	Andhra
Ramu	1450	Andhra
Kamala	2300	Andhra
Sekar	600	Andhra
Sugana	1200	Andhra
Venu	1150	Andhra
Sai	500	Andhra
Shanthi	320	Andhra
+-----+-----+-----+		
80 rows in set (0.00 sec)		

RESULT:

Thus, the JOIN operations – Equi Join, Inner Join, and Outer Join were successfully performed on the given relations in the milk_dairys database. The related records from multiple tables were retrieved successfully, and the outputs of all JOIN operations were verified.